3 · 12

12

· · —

= = =

! (3)

- ( ) . .
- 2 . !
- 1 . — !

" !"

— 3

1. 1 — : · ? (, , DNA)
2. 2 — : ? (, , HR)
3. 3 — : , !

1 — : LEGO +

1) ?

( ) . 46 . (23 + 23)

= ( ) / = 46

1 (1)

... 46
(23 + 23

= DNA
ATCG
ATTCGA...
=
: " !"

, ,
!

1 — (46) DNA( ) ( )

: = , DNA= , = (: " "). 46 !

2) — ?

150 . / ( ) . “ ,
” .

( )

/ — !
/ , /
!

—
, , ...

2 — “ ”

/ , /
.

.

3) —

(Rr) (Rr) . !

Rr × Rr ? (R= , r=)

\	R	r
R	RR	Rr
r	Rr	rr

: 3 : 1
4 3 !
RR 1 , Rr 2 , rr 1
3/4, 1/4
RR 1/3!

3 — : 1 × 1

: R /r 1 , 1 1 . **RR/Rr/Rr/rr 4** !

4) ?

X , **X** **Y** . X (XX), Y (XY) . ! ( )

XX
: X 100%

XY
: X 50% / Y 50%

=

XX / XY
50% / 50%

1

Q. , ?

: “ ” .

▶

5) — !

. , . ,
. . : , !

2 — : +

1) = !

. , . (=) !

(O B A F G K M)

30,000 ! 3,000

5,800

4 — : = , =

: . , . !

2) 2 — vs

. !

( )

( )

(10pc)

! 1 2 . .

: A B 1

A M -5 !

B M 2

3) — ( )

, ? . 1 7 .
!



: (pc) = 1 ÷ (″) 0.1″ 10pc!

5 —

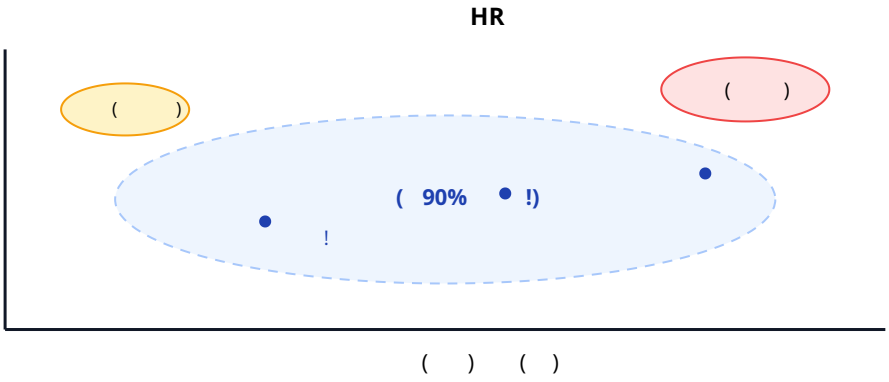
1

Q. 0.1″ 0.01″ ?

▶

4) — HR

HR . ( ), ( ). . !



6 — HR : ,

: = ( ), = ( ), = ( )

5) —

( ): ( ) ( ) ( ) ( )
(8): ! /

100 |

( )

1

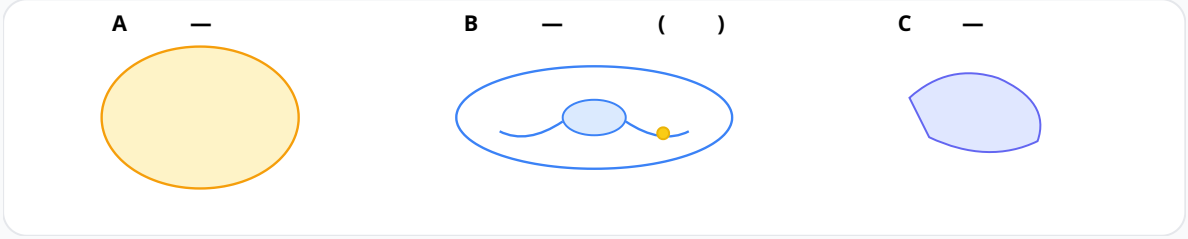
Q. **(:)** **?**

▶

3 — : +

1) =

1000 . ( ), ( ) . ( )
. !

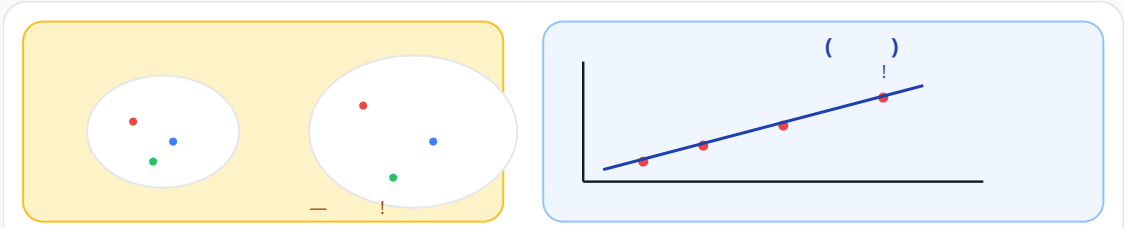


7 — 3 : / /

: = ( ), = (+), = ( )

2) ! —

? . . : $v = H_0 \times d$ (2
2 !)



8 — : ( ), :

) 70, 100Mpc 70×100=7000 km/s . 2% !

: “ ?” **!** . .

3) — (138)

138 . 2 !

1:
( ) !

2:
2.7K !

. .
. 38 138

9 — : .

4) —

100 **5** ! **27(** **) +** **68(** **!)** .
. !

(100)

68 — !

27 **5**

!
“ !”

5 ! 95

5)

“ ” “ ?” , . .
!

“ ?”

. .

. !

10 — : (!)

Q. “ ” ?

▶

? —

__ , DNA __
. __ .

▶

__ , __ , __
.

▶

__ __ , __
. __5 !

▶

?

3 ·
!